

Experiment no: 6 MEASUREMENT OF MECHANICAL PROPERTIES

Introduction

Mechanical properties indicate the response of a metal or alloy to elastic and plastic deformations under the applied forces. There are many tests to determine mechanical properties. Tensile testing is also one of such important tests.

Tensile test

This test is widely used to determine strength, ductility, resilience, toughness and several other material properties. A test specimen of circular or rectangular cross section of a suitable size is prepared from the material to be tested. During the preparation of the specimen sharp changes in section is to be avoided to reduce stress concentration. This reduces the chances of failure at low stress values. The specimen is subjected to increasing tensile loading until it fractures. A record of load acting on the specimen with progressive extension of the specimen is obtained. A schematic illustration of the typical tension specimen is shown in Fig. 1.1

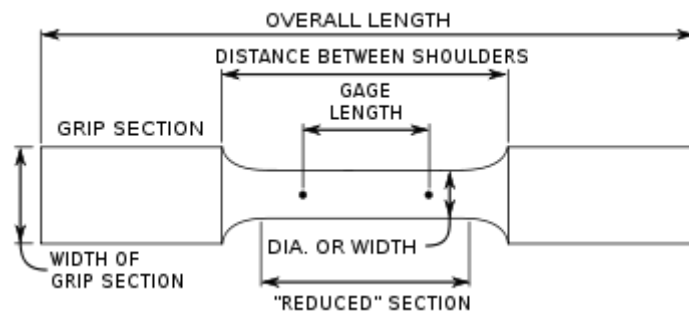


Fig. 1.1 Typical tension specimen

Engineering (i.e. Nominal or Conventional) STRESS-STRAIN curves

These curves are obtained from load-extension curves on the assumption that the dimensions of sample does not change during testing.

Engineering stress (σ_E) = Applied load (P) / original cross-sectional area (A_0)

Engineering strain (e) = Change in gauge length ($d l$) / original gauge length (l_0)

Engineering stress-strain diagram will be same as load-extension diagram. This is due to the fact that if the parameters on X and Y axis are divided by certain constants, the original diagram remains unaffected.

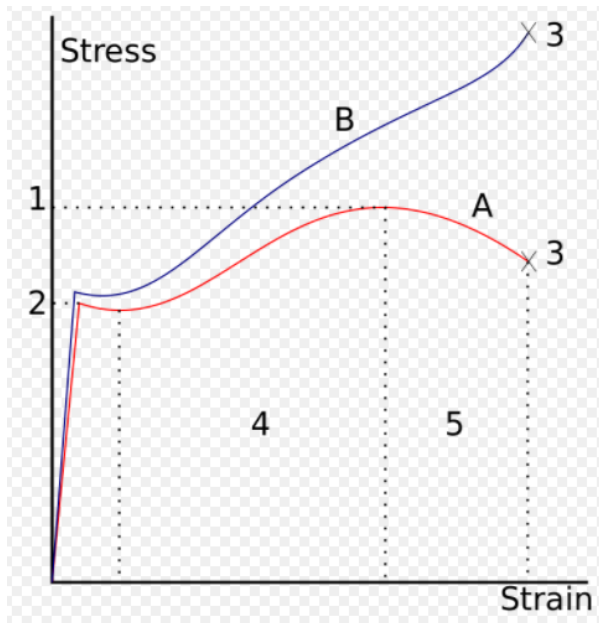
True STRESS-STRAIN curves

True stress (σ_T) = Instantaneous load (P_i) / Actual cross-sectional area at that instant (A_i)

True strain (ϵ) is the summation of all the engineering strains from instant to instant.

$$\epsilon = \int_{l=l_0}^{l=l_i} dl/l$$

A schematic of the engineering and true stress-strain curves is shown in Fig.1.2



Curve A : Engineering Stress Strain Curve

Curve B : True Stress Strain Curve

Fig. 1.2 Engineering and true stress-strain curves

Relationships between engineering and true stress strain values

$$\sigma_T = \sigma_E (1+e)$$

$$\epsilon = \ln (1+e)$$

Evaluation of properties

1. Yield Stress

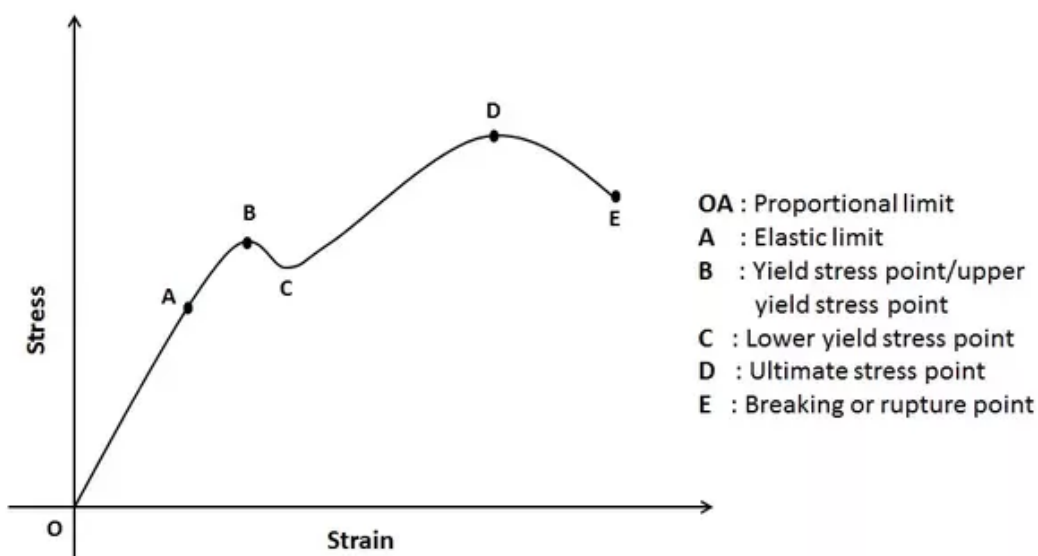
It is the stress at which the material yields i.e. it shows appreciable plastic deformation at almost constant stress without any strain hardening. This stress exists in some materials like low carbon steels and mild steel. It's shown in Fig 1.3 by Point B(Upper yield stress) and Point C(Lower Yield stress).

2. Ultimate Tensile Stress

It is the highest value of the engineering stress that the material can sustain without fracture. It's shown in Fig 1.3 by point D.

3. Breaking Stress or Fracture Stress

It is the stress value at the point of fracture as shown in fig 1.3 by Point E.



OA : Proportional limit
A : Elastic limit
B : Yield stress point/upper yield stress point
C : Lower yield stress point
D : Ultimate stress point
E : Breaking or rupture point

Fig.1.3 Engineering stress strain curve showing different properties

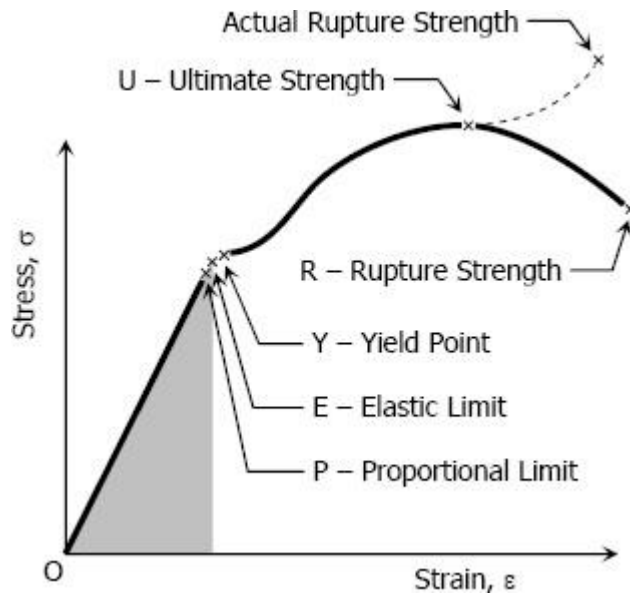
4. Resilience

It is the total energy absorbed by the material during its elastic deformation. It is the area up to the elastic load in a load extension diagram.

Modulus of resilience is the energy absorbed by the material in the elastic region per unit volume.

$$\text{Modulus of resilience} = \frac{\sigma^2}{2E}$$

Where σ is the elastic stress and E is the young's modulus of elasticity.



Shaded Area in the figure 1.4 shows Modulus of Resilience

Fig.1.4 Engineering stress-strain diagram showing modulus of resilience

5. Toughness

It is the total energy absorbed by the material prior to its fracture. It is the total area under the load extension diagram. Modulus of toughness is the energy absorbed by the material prior to its fracture per unit volume. It is the total area under the stress-strain diagram as shown in the Fig.1.5.

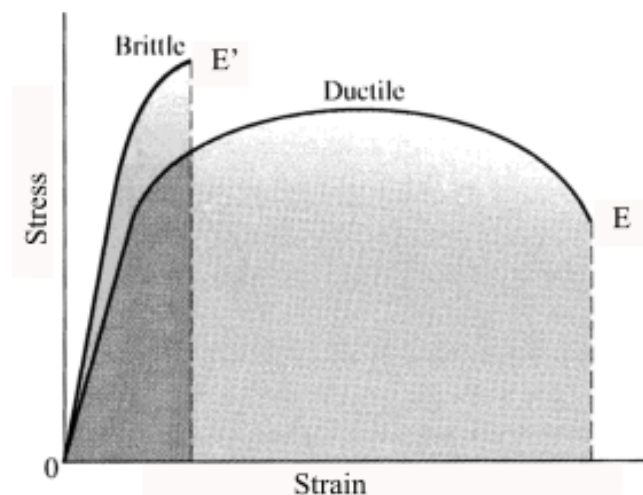


Fig.1.5 Engineering stress-strain diagram showing modulus of toughness

Modulus of toughness can be approximately found out by using the following equations.

$$\begin{aligned}\text{Modulus of toughness} &= ((\sigma_1 + \sigma_u) \times e_f) / 2 \\ &= 2/3 \sigma_u \times e_f\end{aligned}$$

Where σ_1 = Elastic Limit Stress

σ_u = Ultimate Tensile stress

e_f = strain at the fracture point

6. Stiffness

It is the resistance of a material for elastic deformation and is expressed by young's modulus of elasticity.

$$\text{Young's modulus of elasticity (E)} = \text{True stress } (\sigma_T) / \text{True strain } (\epsilon)$$

where both the values of σ_T and ϵ are from the elastic region.

It is the slope of the initial linear part of true stress-strain curve as shown in the Fig.1.6.

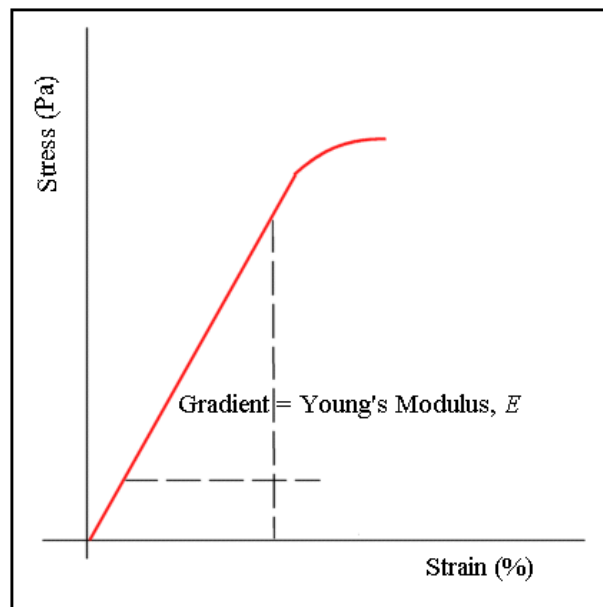


Fig.1.6 True stress-strain curve showing young's modulus

7. Ductility

It is the ability of a material to exhibit large plastic deformation prior to fracture under tensile loading conditions. It is expressed by percentage elongation.

$$\% \text{ Elongation} = \frac{\text{Final length}(l_f) - \text{original length}(l_o)}{\text{original length}(l_o)} \times 100$$

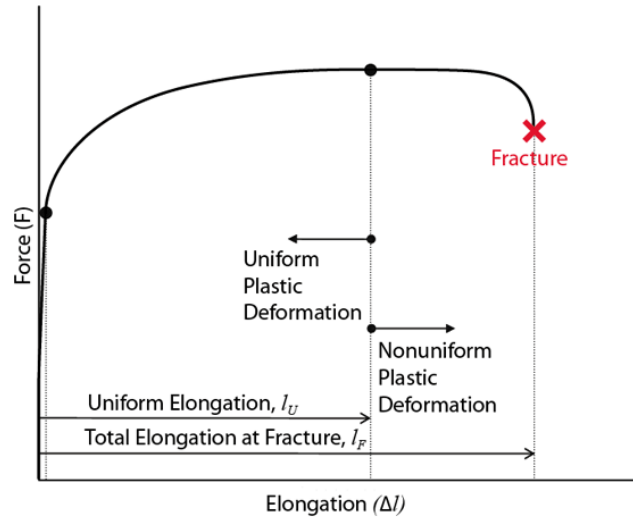


Fig.1.7 A typical load extension diagram showing ductility

8. Strain hardening exponent or coefficient (n)

This coefficient gives an idea about the ability of a material for hardening due to straining or plastic deformation. The approximate equation applicable for deformation behaviour in the plastic region up to necking is

$$\sigma_T = K \epsilon^n$$

where σ_T = true stress,

ϵ = true strain,

n = strain hardening coefficient and

K = material constant called the strength coefficient

A typical true stress strain curve and variation of $\log \sigma_T$ and $\log \epsilon$ is shown in the Fig.1.8 (a) and Fig.1.8 (b) respectively.

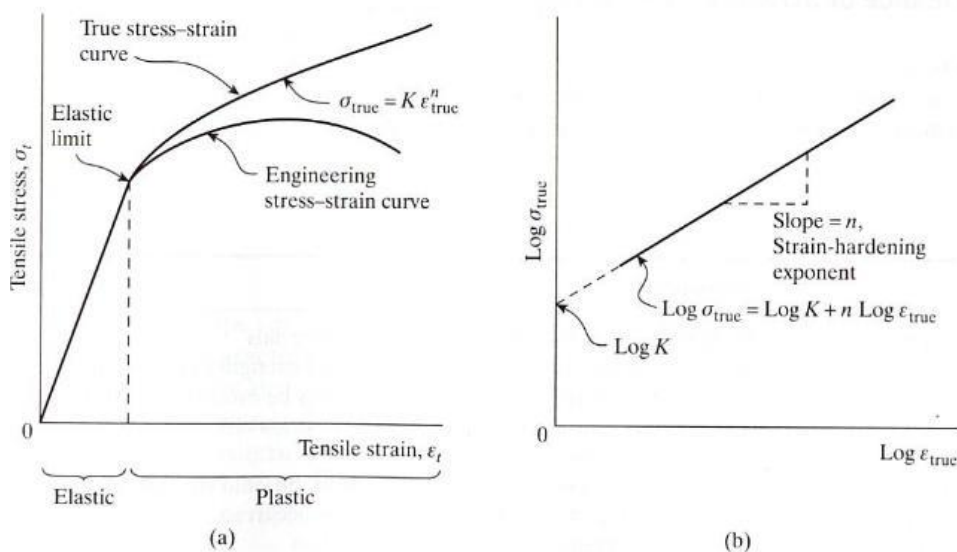


Fig.1.8 A typical true stress strain curve and variation of $\log \sigma_T$ and $\log \epsilon$